

Algorithm Analysis And Data Structures

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Binary Search

Binary search. Given value and sorted array $a[]$, find index i such that $a[i] = \text{value}$, or report that no such index exists.

Invariant. Algorithm maintains $a[\text{lo}] \leq \text{value} \leq a[\text{hi}]$.

Ex. Binary search for 33.

6	13	14	25	33	43	51	53	64	72	84	93	95	96	97
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
↑														↑
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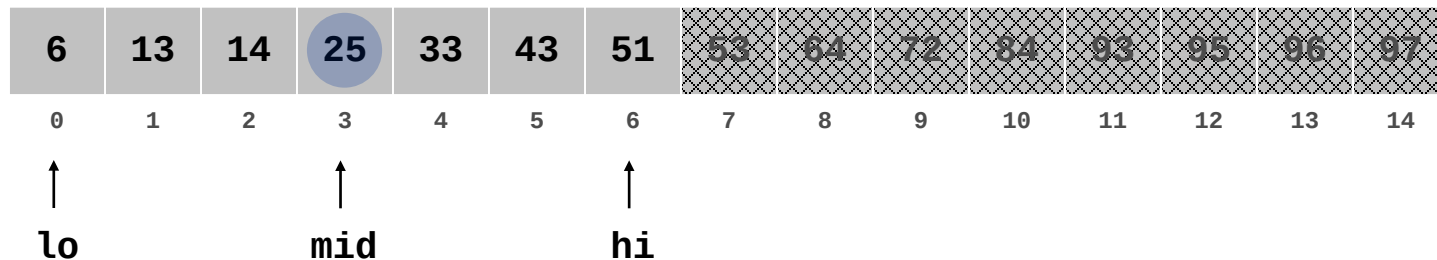
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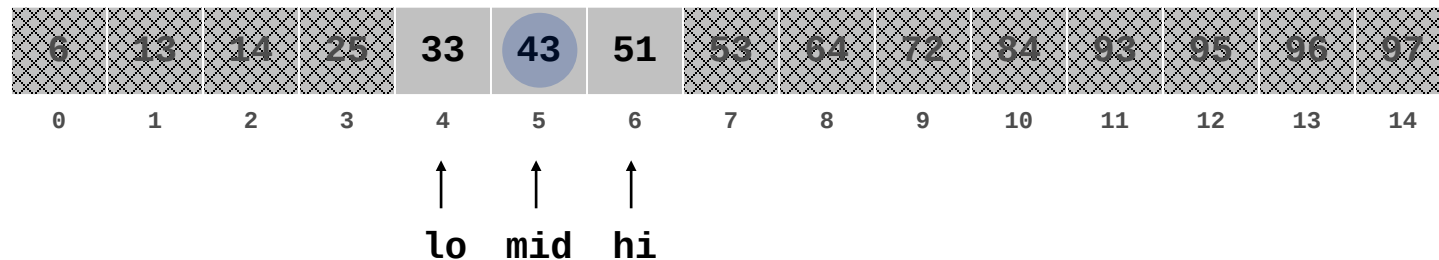
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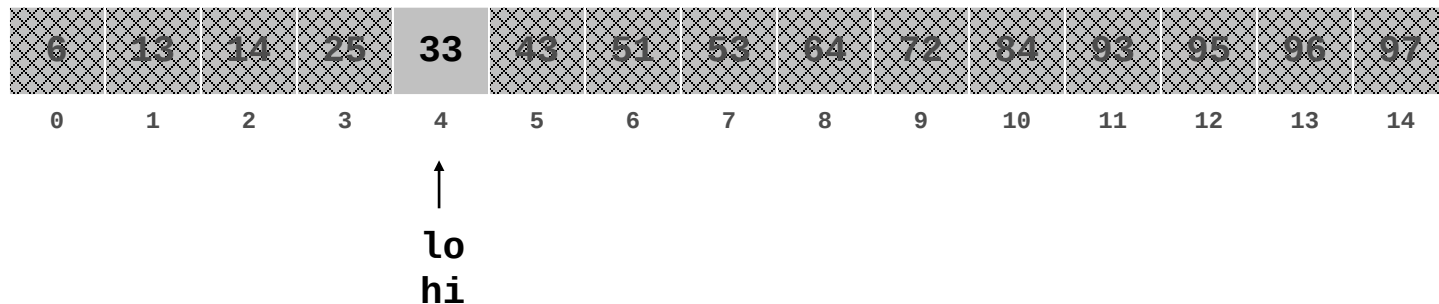


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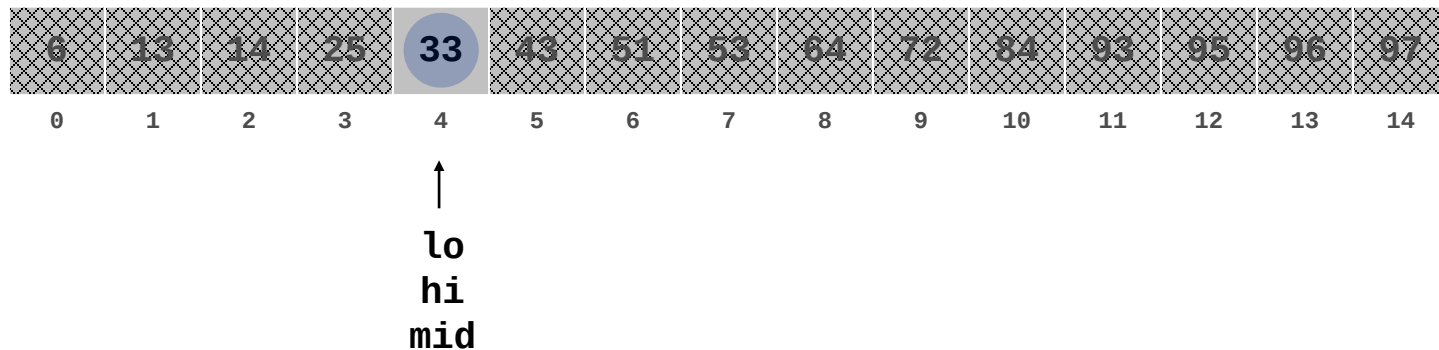


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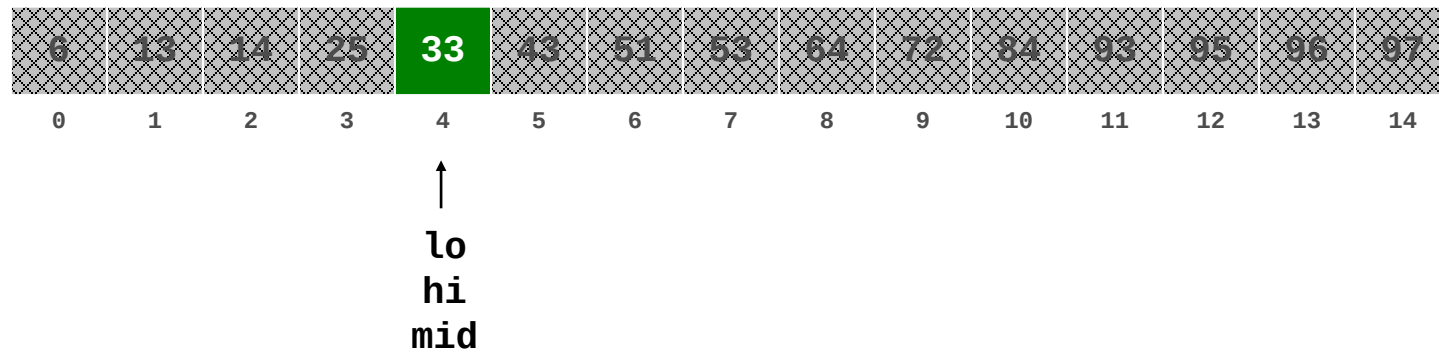


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Binary Search – Analysis

For an unsuccessful search:

- The number of iterations in the loop is $\lfloor \log_2 n \rfloor + 1$
→ $O(\log_2 n)$

For a successful search:

- **Best-Case:** The number of iterations is 1. → $O(1)$
- **Worst-Case:** The number of iterations is $\lfloor \log_2 n \rfloor + 1$ → $O(\log_2 n)$
- **Average-Case:** The avg. # of iterations $< \log_2 n$ → $O(\log_2 n)$

0	1	2	3	4	5	6	7	← an array with size 8
3	2	3	1	3	2	3	4	← # of iterations

The average # of iterations = $21/8 < \log_2 8$

How much better is $O(\log_2 n)$?

<u>n</u>	<u>$O(\log_2 n)$</u>
16	4
64	6
256	8
1024 (1KB)	10
16,384	14
131,072	17
262,144	18
524,288	19
1,048,576 (1MB)	20
1,073,741,824 (1GB)	30

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